

Topics

o Heine-Borel

o Limit point compactness

o Cantor set

Recap

Examples:

<u>Compact</u>	<u>not compact</u>
finite metric spaces	$\mathbb{N}$
$\{\frac{1}{n} n \in \mathbb{N}\} \cup \{0\}$	$\{\frac{1}{n} n \in \mathbb{N}\}$
d -cells in $\mathbb{R}^d$ Ex: $[a, b] \subset \mathbb{R}$	

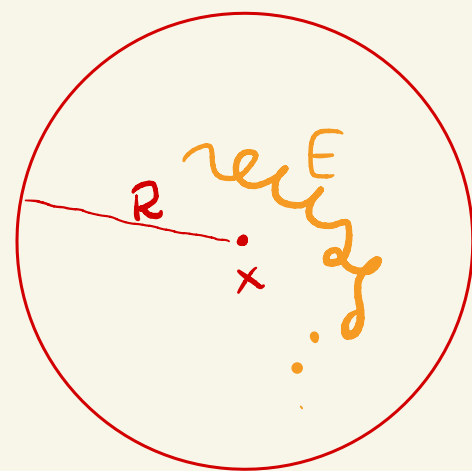
Results:

- Compact subsets are closed
- Closed subset of compact is compact
- Infinite subset of cpt set has limit point

Def A subset E of a metric space X is

bounded if there exists $x \in X$ and $R > 0$

such that $d(x, p) < R$ for all $p \in E$.



Remark: True for some x
 $\Rightarrow$ true for any x .

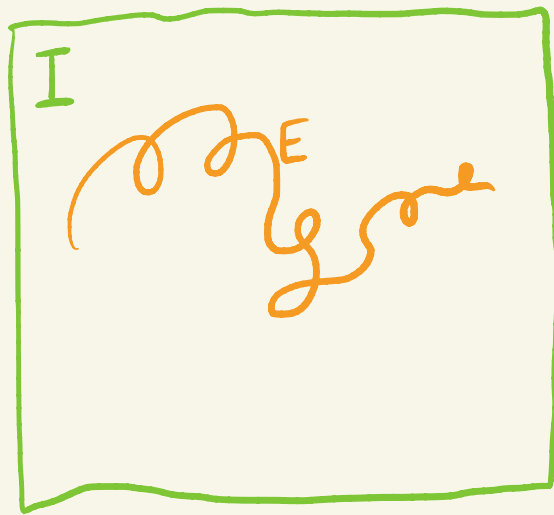
Thm (Heine-Borel) The following are equivalent for $E \subset \mathbb{R}^d$.

(a) E closed and bounded

(b) E compact

Proof (a) $\Rightarrow$ (b)

Suppose E closed + bounded



Since E is bounded, it is a subset of some d -cell.

Closed subsets of compact sets are compact, so E is compact.

(b) $\Rightarrow$ (a)

Suppose E is compact.

boundedness

Let $p \in X$. Let $U_r = N_r(p)$.

$\{U_r\}_{r \in \mathbb{R}_{>0}}$ is an open cover of E .

$\Rightarrow \exists r_1, \dots, r_n$ s.t. $U_{r_1} \cup \dots \cup U_{r_n} = U_{r_{\max}}$ contains E .

$\Rightarrow E$ bounded.

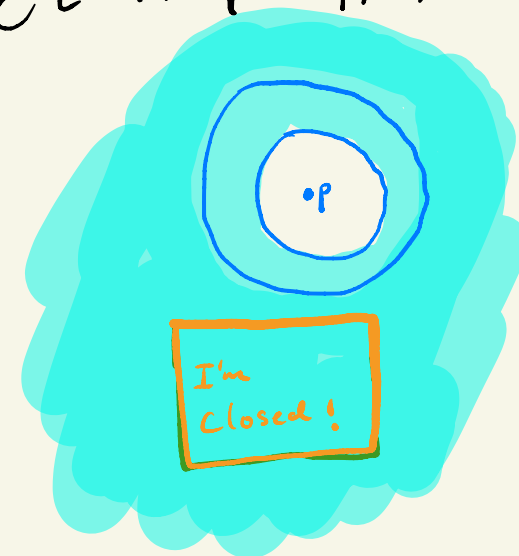
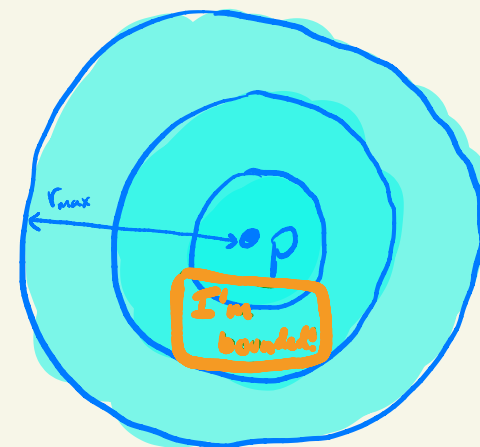
closedness

We show E^c is open. Let $p \in E^c$. Let $V_r = \{q \in X \mid d(p, q) > r\}$.

$\{V_r\}_{r \in \mathbb{R}_{>0}}$ is an open cover of E .

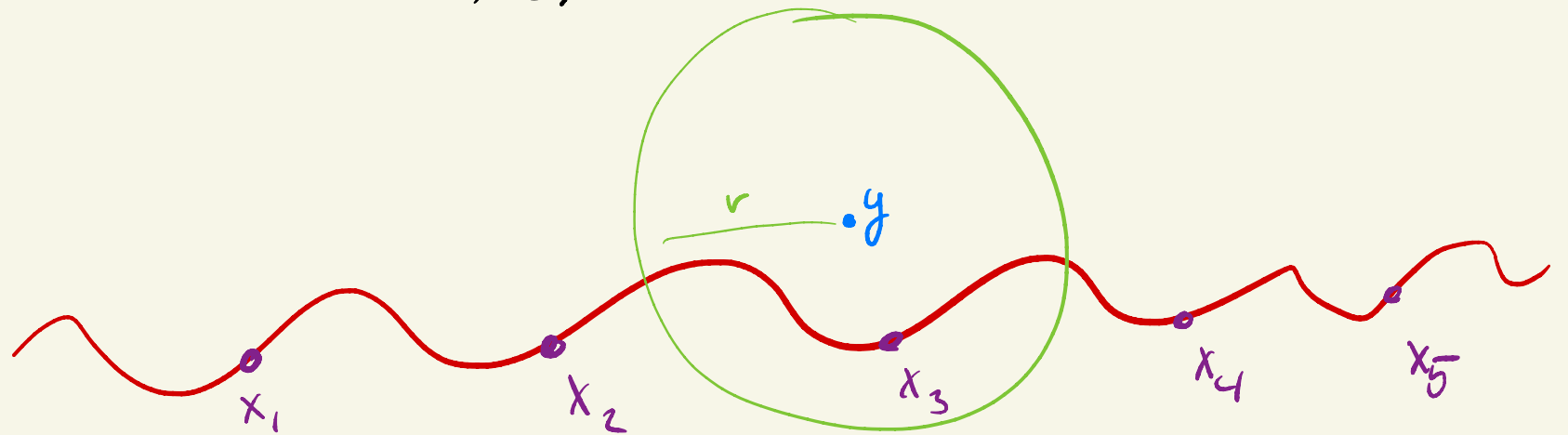
$\exists r_1, \dots, r_n$ s.t. $V_{r_1} \cup \dots \cup V_{r_n} = V_{r_{\min}}$ contains E .

$\Rightarrow N_{r_{\min}}(p)$ is in E^c .



Thm $E \subset \mathbb{R}^d$ is compact iff every infinite subset of E has a limit pt in E .

Suppose E not bounded. There is a seq $x_1, x_2, x_3, \dots$ in E
with $|x_n| > n$.



Claim: $\{x_i\}$ is an infinite subset of E with no limit point.

Let $y \in X$. Suppose $d(y, x_n) < r$.

$$d(y, x_n) + d(y, 0) \geq d(x_n, 0)$$

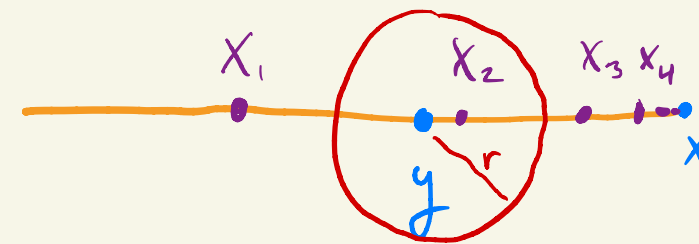
$$r + d(y, 0) \geq n$$

This holds for finitely many n , so y not limit point.

Suppose E not closed. Let $x \in E' \setminus E$. Let $x_n \in E$ be s.t. $|x_n - x| < \frac{1}{n}$

Claim: $\{x_i\}$ is an infinite subset with no limit point in E .

Suppose $y \in E$



Let $0 < r < d(y, x)$. Suppose $d(y, x_n) < r$.

$$d(y, x) \leq d(y, x_n) + d(x, x_n) < r + \frac{1}{n}$$

$$d(y, x) - r < \frac{1}{n}$$

$\frac{1}{d(y, x) - r} > n$. Only finitely many such $n \Rightarrow y$ not limit point.

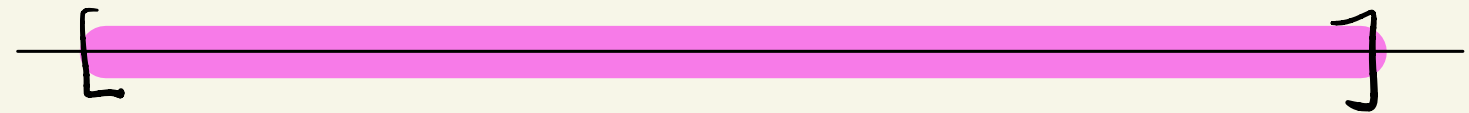
Rmks

- In any metric space, limit pt. compactness $\Leftrightarrow$ compactness (Rudin exercise 26)
- Heine-Borel is not true in general metric spaces
 - $[a,b] \cap \mathbb{Q}$ is closed + bounded in $\mathbb{Q}$, but not compact.
 - Infinite set with $d(x,y) = \begin{cases} 0 & x=y \\ 1 & x \neq y \end{cases}$
 - The closed ball $\{x \in V \mid \|x\| \leq 1\}$ is not compact in an infinite dimensional vector space with any reasonable metric, as $(1, 0, 0, \dots), (0, 1, 0, \dots), (0, 0, 1, \dots)$ has no limit points.

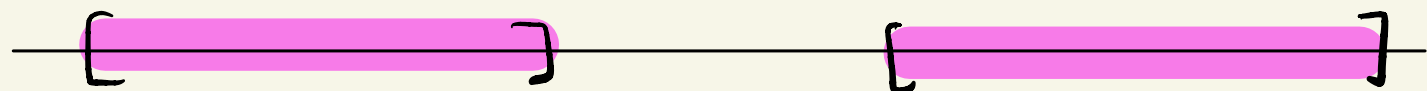
The Cantor set

Q Does every closed, uncountable $E \subset \mathbb{R}$ contain an interval?

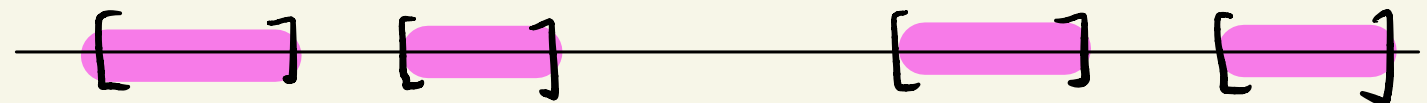
$$E_0 = [0, 1]$$



$$E_1 = [0, \frac{1}{3}] \cup [\frac{2}{3}, 1]$$



$$E_2 = [0, \frac{1}{9}] \cup [\frac{2}{9}, \frac{3}{9}] \cup [\frac{6}{9}, \frac{7}{9}] \cup [\frac{8}{9}, 1]$$



$E_{k+1} = E_k$ minus middle 3rd of each segment

E_n = union of 2^n closed intervals of length $\frac{1}{3^n}$.

Total length of $E_n = \left(\frac{2}{3}\right)^n$, which goes to 0.

Let $C = \bigcap_{n=1}^{\infty} E_n$

- Closed? Yes. intersection of closed $\Rightarrow$ closed.
- compact? Yes. Also bounded.
- Empty? No. intersection of nested compacts.

Also, look at interval endpoints.

- Does C contain an open interval?

Consider (a, b) . E_k cannot contain (a, b)

if $\frac{1}{3^k} < b - a$.

- Countable? No.

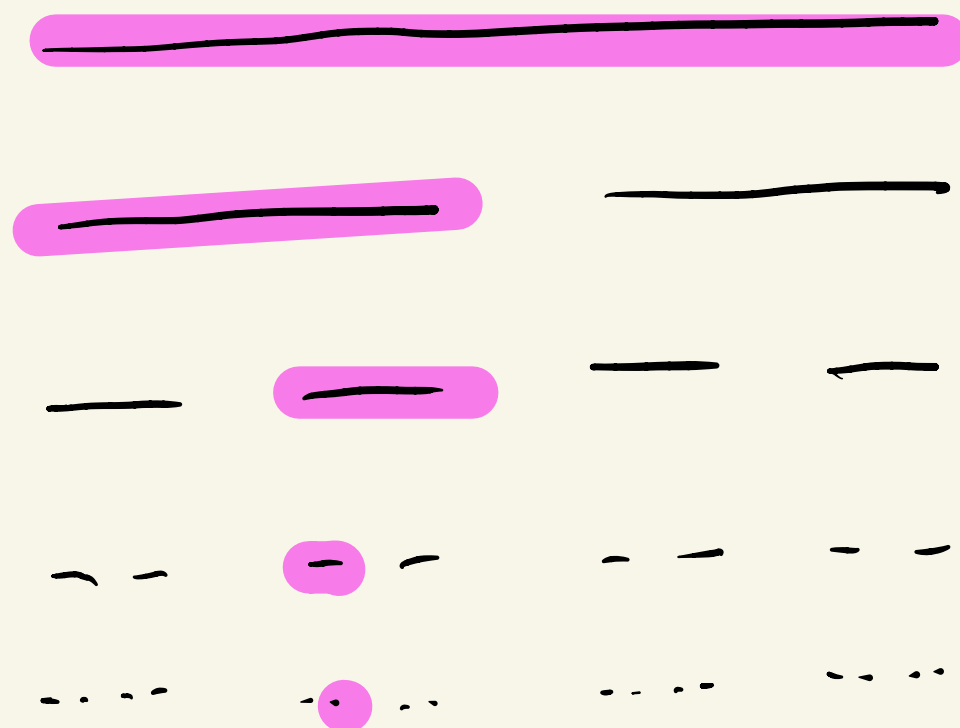
The Cantor set has a bijection with $\{0, 1\}^{\mathbb{N}}$.

A binary sequence (b_k) specifies a sequence of intervals (I_k) where $I_0 = (0, 1)$ and

I_k is the left $\frac{1}{3}$ of I_{k-1} if $b_k = 0$

and I_{k-1} is the right $\frac{1}{3}$ if $b_k = 1$.

$0, 1, 0, 1, 1, \dots \mapsto$



We map the seq. (b_i) to the single point in $\bigcap_{k \in \mathbb{N}} I_k$.